Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/26/2007
PI Name:	Lackner, James R. Ph.D.		
Project Title:	Coordination of Turning and Reaching		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02454-9110	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
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No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 12/2/2006 per J. Dardano (4/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	DiZio, Paul (Brandeis University)		
Grant/Contract No.:	NAG9-1466		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We have developed a turn and reach paradigm to assess post-flight disturbances of arm-trunk coordination. Our recent ground-based work has shown that simultaneous turn and reach movements involve high rotational velocities of the trunk that generate large, potentially disrupting Coriolis forces on the reaching arm and on balance. Nevertheless, reaching accuracy is preserved because the central nervous system programs compensation for the self-generated Coriolis forces. Failure to predict accurately Coriolis force magnitude produces disruption of coordination and potentially loss of balance. Our aims are: Aim 1-Parametrically evaluate Coriolis force influences on kinematics and dynamics of voluntary turning and reaching to targets under conditions of normal and increased body weight. Aim 2-Measure reaching movements and postural control after motor adaptation to rearrangement of the relationship between voluntary torso rotation relative to the feet and inertial torso rotation in space. Aim 3-Assess whether motor adaptation to Coriolis force perturbed reaching acquired in a rotating room transfers to turn and reach movements in a natural environment. These		

	experiments will be of value in pinpointing factors contributing to post-flight postural and arm-trunk coordination deficits. They will also have significance for understanding potentially contributing factors in populations at risk for falling.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	How the neuromotor system controls the reactive forces that arise between moving segments of the arm has been studied for years, because several patient populations, including stroke patients and Parkinson's patients have been shown to suffer from specific deficits in control of interaction forces among limb segments. Our previous studies in a rotating artificial gravity environment opened a new perspective on the issue which was relevant for both astronauts and terrestrial patients. The new finding was that Coriolis forces generated between the arm and torso during everyday turning and reaching movements are much larger than interaction forces between arm segments, yet we can adapt to them rapidly and completely. Thus, this project aims to understand the learning process which lets us move our arms accurately without having to keep our body perfectly still and which helps us avoid loss of balance when moving our arms.
Task Progress:	The overall goal of this project is to study reaching movements made during simultaneous voluntary torso rotations, which we call turn and reach movements (T&R). These movements generate Coriolis forces which are large enough to cause substantial reaching errors, but such errors do not occur. Understanding the process of compensation for T&R is tantamount to understanding motor learning in novel force environments. During this reporting period, which covers the termination of this funded project, we finished the two specific aims that had not been accomplished in previous periods. The first experiment demonstrated that reaching movements are altered following experimental induction of altered control of the torso. The nature of the alterations narrows down the likely processes by which we compensate for self-generated loads on our movements. The second experiment demonstrated that compensation for externally generated loads during T&R is nearly optimal. This reveals that humans can segregate internal and external loads and generate appropriate predictive compensations for each component. The first experiment was motivated to understand the process by which we compensate for Coriolis forces generated on the moving arm by simultaneous voluntary torso rotation. We wanted to distinguish between two alternative mechanisms: either a predictive feed forward muscle activation profile is generated on the basis of an internal estimate of the forthcoming combination of torso angular velocity relative to space and arm linear velocity relative to the torso, or reactive feedback compensation is produced due to spinal stretch reflexes and vestibulospinal reflexes. The experimental rationale was to distort motor control of torso rotary movements and then measure whether arm movements would consequently be distorted. If arm movements were normal then accurate feedback about the abnormal torso rotation would have to be considered responsible for correcting it, but if the arm movements showed directional then an improper feed forward compensation based on planned but erroneously executed torso rotation would be the most likely mechanism. We altered motor control of torso rotation movements by having subjects swivel their torso while standing on a platform programmed to swivel at a proportion of the angular speed of their torso (relative to their feet). (We had developed this procedure for altering motor control of the torso in previous project years.) If the floor turned in the same direction as their torso, it created a situation where the motion of the torso relative to space was larger than the angular rotation of the torso relative to the feet. Inertial motion governs the torque required to move, so in this situation subjects had to learn to increase their foot-to-torso rotary torques in order to achieve a constant rotary excursion of their torso relative to the feet. This was done in darkness, and subjects did not detect that their inertial motion had changed. After many leftward torso movements (without moving the arms) during progressively larger leftward floor rotations, we stopped the floor and had subjects make either additional torso movements without arm movements, arm movements without torso movements, or T&R movements. The post-adaptation spatial extent and speed of torso movements were amplified relative to baseline movements. After adaptive potentiation of leftward torso rotation the finger velocity relative to the torso was the same as pre-adaptation, so the rightward Coriolis force was increased. During post-adaptive T&R movements, there was a radial deviation of the arm final position and peak velocity relative to the torso, rightward of the pre-adaptation baseline. Arm movements made without torso rotation did not differ before and after adaptive recalibration of torso movements. The arm deviations seen in the T&R movements indicate subjects had pre-programmed their reaches to compensate for the rightward Coriolis force that would be present if they physically turned leftward the amount they intended rather than the actual larger amount. Since astronauts underestimate their turns after reentry we expect their whole body movements would be deranged until re-adaptation takes place. The T&R paradigm would allow a comprehensive assessment of recovery, and the paradigm of torso-contingent floor rotation could hasten re-adaptation or provide pre-adaptation. The second set of experiments we completed evaluated the effects of the background force environment and of hefted loads on compensation for Coriolis forces in T&R movements. One experiment examined T&R movements under conditions of normal, decreased and increased body weight, in parabolic flight. The experimental results demonstrate that under 1g conditions subjects when turning and reaching for a target are very accurate, regardless whether they reach slow or fast, or without visual feedback. Normal subjects show temporally overlapping arm and torso velocities, with the peak velocities occurring virtually simultaneously. As a consequence, very large Coriolis forces are generated on the reaching arm; nevertheless these forces do not disrupt movement accuracy. By contrast, reaching movements during passive body rotation have large trajectory deviations and endpoint errors. These results confirm that in natural turn and reach movements there is an anticipatory feed forward compensation for the impending Coriolis force which maintains straight and accurate reaches in external space coordinates. Subjects exposed to 0 g (feet anchored to the floor) and 1.8g force levels in parabolic flight initially exhibit disruptions of movement paths and endpoints. Within 40-50 turn and reach movements they achieve a great deal of compensation. This indicates that the T&R paradigm is useful for evaluating whole body performance in non-1g environments and for pre-adapting subjects who are about to undergo force transitions (e.g., re-entry to Earth gravity, transition from weightlessness to artificial gravity). Another experiment explored the effects of hefted loads on T&R movements. Subjects made T&R movements while grasping objects weighing 500g or 1000g. In addition to providing a static vertical load, the objects increased the dynamic Coriolis force which needed to be compensated during the T&R movements. We found the compensation was optimal under both loads conditions as well as in unloaded conditions. This was true when subjects used their preferred or non-preferred hand. The results indicate the humans can generate automatic compensations for impending Coriolis

	forces based on estimates of their own body dynamics and the dynamics of a hefted object. A final technical accomplishment was the development of a mathematical model that allows us to compute the inverse dynamics of turn and reach movements, movements made in a rotating reference frame, and movements perturbed by an external mechanical load.
Bibliography Type:	Description: (Last Updated: 03/31/2017)
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